

Timeflow Test Automation Framework

Comprehensive Technical & Architectural Engineering Report | Setoo QA Automation

Target Application:	https://timeflow.setoo.in	Suite Pass Rate:	100% (52/52 Tests Passed)
Core Language:	Java 17 (OpenJDK LTS)	Automation Core:	Selenium WebDriver 4.29.0
Test Runner:	TestNG 7.10.2	Architecture Pattern:	Page Object Model (POM)
GitHub Repository:	github.com/darshanhujband8055/timeflow-selenium-automation	ALM / Project:	Azure DevOps (setoo-ai / Timeflow)

1. Executive Summary

The Timeflow Selenium Automation Framework is an enterprise-grade automated regression and quality audit solution designed to validate end-to-end user workflows, access security, and business invariants across all 4 user personas on the Setoo Timeflow web application. The framework was engineered with a primary focus on **zero-friction portability**: any developer, QA engineer, or manager can clone the repository and execute all 52 tests locally via a 1-click batch runner without manually configuring build runtimes or downloading browser binaries.

2. Tools, Technologies & Services Stack

Category	Technology / Tool	Version / Role	Technical Significance
Language	Java	17 (LTS)	Provides strong static typing, OOP reliability, and enterprise stability.
Core Automation	Selenium WebDriver	4.29.0	Automates Chrome, Edge, and Firefox. Bundled Selenium Manager manages driver binaries automatically.
Test Runner	TestNG	7.10.2	Drives test suite orchestration, lifecycle hooks, assertions, and test listeners.
Build Automation	Apache Maven	3.9+ & mvnw	Build lifecycle, dependency management, and Surefire execution reporting.
Target Frontend	Next.js / React	SPA / Radix UI	Modern web app with async hydration, interactive tab navigation, and client modals.
Target Backend	PostgreSQL / Node.js	Supabase / REST	Relational data store persisting timesheets, project budgets, and user entitlements.
Authentication	JWT / OAuth	Session Guard	Role-gated sessions across Employee, Manager, PM, and Administrator.
Version Control	Git & GitHub	main branch	Centralized repository with automated wrapper scripts and documentation.
ALM Integration	Azure DevOps	setoo-ai	Connected to project 'Timeflow timesheet' for work item alignment and bug tracking.

3. Framework Architecture & Design Pattern

The framework strictly adheres to the **Page Object Model (POM)** architecture, decoupling UI element locators, user actions, and test verification logic into discrete maintainable layers:

- Page Object Layer** (`src/main/java/com/timeflow/pages``): Encapsulates page elements (`private final By ...``) and user actions in dedicated classes (e.g. `EmployeeTimesheetsPage``, `PMBillingPage``, `ManagerApprovalsPage``). Locators are isolated so UI changes require updates in only one file.
- ThreadLocal Driver Management** (`DriverManager.java``): Wraps WebDriver in a `ThreadLocal`` container to ensure thread safety, complete test isolation, and seamless multi-threaded parallel execution.

- **Dynamic Synchronization (`BasePage.java`):** Replaces brittle hardcoded sleeps with intelligent explicit synchronization using `WebDriverWait` and `ExpectedConditions` (`elementToBeClickable`, `visibilityOf`, `invisibilityOfElementLocated`).
- **Externalized Configuration (`config.properties`):** Allows dynamic runtime toggling of target browser (chrome/edge/firefox), application URL, headless mode (`-Dheadless=true`), and timeout thresholds without modifying source code.

4. Execution Workflow & Zero-Setup Portability

Execution Method	Command / Action	Target Audience & Workflow
1-Click Windows Runner	Double-click run-all-tests.bat	Ideal for managers. Bundled Maven Wrapper handles setup automatically and launches the full suite in Google Chrome.
1-Click Mac/Linux Runner	./run-all-tests.sh	Shell script for macOS and Linux workstations with execution permissions pre-configured.
Maven CLI (Full Suite)	./mvnw clean test	Executes all 52 tests mapped in testng.xml across all 4 business modules.
Headless Mode (CI/CD)	./mvnw test -Dheadless=true	Runs silently in the background without UI popups—ready for Azure DevOps CI pipelines.
Module-Specific Run	./mvnw test -Dtest=ManagerModuleTest	Selectively executes only the tests for a target module or role.
IDE Execution	Right-click testng.xml -> Run	For developers and SDETs working in IntelliJ IDEA, VS Code, or Eclipse.

5. Failure Handling & Diagnostic Strategy

- **Automated Failure Screenshots:** Integrated via TestNG's `ITestListener` (`TestListener.java`). Any unexpected assertion or timeout failure automatically captures a full-screen timestamped PNG screenshot saved directly to `target/screenshots/`.
- **Surefire HTML Reports:** Generated at `target/surefire-reports/index.html` detailing test class run times, method parameters, failure traces, and aggregate pass rates.
- **Driver Auto-Cleanup:** The `@AfterMethod` lifecycle hook guarantees that browser windows and driver instances terminate cleanly even if assertions fail, preventing orphaned processes from leaking machine memory.

6. Module-by-Module Automation Script Breakdown

Module A: Standard Employee shubham.shinde@setoo.co / Pass@123	Test Classes: EmployeeModuleTest, EmployeeTimesheetWorkflowTest, EmployeeAuthenticationTest, EmployeePermissionsTest Scope: Daily/Weekly/Monthly timesheet toggles; 'Sync My Tasks' task ingestion; hours logging & dynamic sum calculations; 'Save draft' persistence; 'Submit week' modal guardrails; navigation to My Tasks, Allocations, and Calendar; RBAC negative tests blocking admin routes.
Module B: Reporting Manager rutuja@setoo.co / Pass@123	Test Classes: ManagerModuleTest Scope: Approvals queue processing; table checkbox multi-select; batch approval workflows; batch rejection workflows; rejection feedback comment enforcement (DEF-MGR-001 audit); Team Directory member profiles; Bandwidth allocation gauges; capacity utilization thresholds; date filters and CSV exports.
Module C: Project Manager (PM) rohan@setoo.co / Pass@123	Test Classes: PMModuleTest Scope: Project portfolio overview; project health status badges; sprint velocity metrics; Billing & Budget dashboard (/billing); financial stat cards (Planned vs. Consumed hours, Revenue); Extra Burned calculation verification (DEF-PM-001 audit); mutation authorization guardrails (TC 1.3).
Module D: Administrator darshan@setoo.co / Pass@123	Test Classes: AdminModuleTest, AdminRejectionWorkflowTest, AccessControlAndPermissionsTest Scope: Full master privilege verification; organizational overview cards (Active Users, Active Projects, ADO Mapped status, Pending Approvals); tab navigation (Overview, Approvals Analytics, Pending Submissions); Organization Settings (/settings); global security RBAC negative tests.

7. Defect Audits & Next Steps

A total of **17 functional defects** across all 4 modules were identified during framework execution and exploratory testing. Each defect has been cataloged with reproduction steps, root cause analysis, and developer remediation code in the module PDF reports. The complete test framework is version-controlled on GitHub at <https://github.com/darshanujband8055/timeflow-selenium-automation>.